DAY 35 | POST-READ

Observability, Drift and Optimisation: Recap and Beyond

What we covered today, plus what to carry into the capstone.

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Part A: Summary of today

Topic	What to remember
Signals	Logs, metrics, traces, plus quality and safety signals.
Structured logs	JSON per line with request id, release, prompt id, model, tokens, latency, context ids.
OpenTelemetry GenAI (NEW v4)	Standard gen_ai.* span attributes; content capture opt-in; check the spec version.
Cost per trace (NEW v4)	Sum tokens across spans of one trace; price with placeholder config; alert on outliers.
Alerts	Sustained windows, relative changes, an owner and a runbook for each.
Feedback loop	Collect, triage, fix, add a golden test, verify with the quality gate.
Drift types	Data, concept, knowledge, provider and model (NEW v4), output.
Detection	PSI, KS test, embedding centroid shift; response model changes.
Scheduled evals (NEW v4)	CI schedules or Codex automations; non-zero exit opens a human review.

Part B: Going beyond today

1. LLM observability tools

Several tools record prompts, retrieved context, outputs, tokens and scores per request, and many now accept OpenTelemetry GenAI spans. Check data handling and retention before sending customer data to any tool.

2. Sampling and privacy

Log metadata for all requests and full content for a masked sample, plus every request with negative feedback. Set retention periods that match the customer's policy and local law.

3. Effect size, not only p-values

With millions of requests, tiny meaningless differences become significant. Look at how big the change is, and confirm with real examples.

4. Topic discovery

Cluster recent questions on embeddings and ask a model to name each cluster. New clusters with high refusal rates show exactly which documents are missing.

Part C: Capstone hand-off

Your capstone (Days 36 to 40, 15 to 21 Dec 2026) should include at least: a golden set, a quality gate, structured logs or OpenTelemetry spans, one dashboard, one drift or quality check, a token and cost report, a vulnerability report, and an `AGENTS.md` in the team repo. Evaluators will look for them.

- Start from `Day36-40-Capstone/00_Capstone_Guide.pdf` and the project briefs in `Day36-40-Capstone/Projects/`.
- Reuse your SmartDesk code: gateway (Day 34), resilient client (Day 33), quality gate (Day 32), guardrails (Day 30).
- Code freeze is Day 39 at 17:00 IST; deliverables are due Day 40 at 09:00 IST.

Part D: Check yourself

- Which gen_ai.* attributes would you set on a chat span?
- How do you compute the cost of one trace?
- How do you read a PSI of 0.3?
- What signals tell you the provider changed the model behind the same name?
- Why should a scheduled eval open a review instead of fixing things automatically?

Part E: Practice ideas for tonight

- Read all six capstone briefs and rank your top two with reasons.
- Draft your capstone's golden set with 10 items.
- List which notebooks from Days 2 to 35 you will reuse.

Coming up: the capstone (Days 36 to 40). Real-world project options, built end to end in teams, evaluated with a shared rubric.

Otsukaresama deshita (お疲れ様でした): thank you for your hard work today